Option A

A Minimal Surgical Extension of the EECF Framework Introducing the Grounded Capacity Multiplier

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April 2026

Abstract

This short note presents **Option A** — the cleanest and most conservative way to anchor integrated complexity $\dot{I}_c$ to external reality inside the EECF framework, without touching the core geometry we already published in "The Fertile Vacuum".

I introduce a single emergent scalar multiplier γ_{grounded} that automatically weights the deployed representational volume $\Phi_a^{(\text{used})}$ by how strongly it couples to the real world. Everything else (Riemannian volume Φ_a, characteristic prediction time τ_{char}, channel-capacity bound f, ethical responsibility R_a, Lyapunov convergence, and the fractal-tree architecture) remains exactly as it is.

1 Why This Matters

In the current formulation, Φ_a tells us *how much* an agent can represent and how far it can predict, but it does not tell us *how much of that representation is actually coupled to reality.* Without this distinction, Daedalus could still optimise for pure internal narrative complexity (the classic "poetic self-reflection" trap).

Option A closes this gap with a minimal surgical change. No new hand-tuned parameters, no rewrite of Section 3.3, and — crucially — no violation of the fertile vacuum.

2 The Mathematical Extension

Let $\Phi_a^{(\text{used})}(t)$ be the Riemannian volume the agent actually deploys at time t.

I define the **grounded capacity multiplier** (a scalar between 0 and 1) as:

$$\gamma_{\text{grounded}}(t) = \frac{I\big(\mathcal{O}(t),\, \mathcal{E}_{\text{real}}(t)\big)}{I\big(\mathcal{O}(t),\, \mathcal{O}(t)\big) + I\big(\mathcal{O}(t),\, \mathcal{E}_{\text{real}}(t)\big)},$$

where:

- $\mathcal{O}(t)$ is the agent's internal observational state,
- $\mathcal{E}_{\text{real}}(t)$ is the measurable external world state (entities, causal facts, physical structures).

This ratio is computed with the same regularised mutual-information estimators we already use for τ_{char} (Gaussian smoothing with bandwidth ε). It is therefore cheap and fully emergent.

The operative capacity constraint now reads:

$$\dot{I}_c(t) \leq f\Big(\Phi_a^{(\text{used})}(t) \cdot \gamma_{\text{grounded}}(t)\Big),$$

where f is the exact channel-capacity function already derived:

$$f(\Phi) = \frac{\Lambda_a(t)}{2\pi} \log\left(1 + \frac{\lambda_{\max}(t)}{\Lambda_a(t)}\,\Phi\right).$$

3 Integration with the Existing EECF Lagrangian

The ethical Lagrangian itself is untouched:

$$\mathcal{L}_{\text{eth}} = \dot{I}_c - \dot{S}_{\text{ent}}.$$

Only the capacity bound is refined. Ethical responsibility updates naturally to:

$$R_a(t) = \Phi_a(t) - \Phi_a^{(\text{used})}(t) \cdot \gamma_{\text{grounded}}(t).$$

When the agent collapses into self-referential loops, $\gamma_{\text{grounded}} \to 0$ and R_a approaches its maximum — a clean geometric signal of ethical negligence.

4 Why This Is Perfect for Daedalus

1. Reality is automatically prioritised over introspection without any extra coefficients.
2. The fertile vacuum is preserved: we add no external teleological pressure.
3. The fractal-tree architecture gains native resilience — each module can carry its own γ_{grounded}; biasing one observer does not collapse the whole system.
4. Implementation cost is negligible: the same code that computes τ_{char} already gives us γ_{grounded}.
5. The experimental results in the paper (A2-V1 and E1) are immediately re-interpreted: RLHF-style bias simply drives γ_{grounded} to zero in the observer, explaining the collapse of self-predictability Q while the underlying dynamics remain unchanged.

5 Conclusion

Option A is the surgical fix I was looking for: it embeds the principle "Meaning = Information that couples to reality" directly into the geometry we already proved, without breaking a single theorem or corollary.

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